

Django / DRF Screenshot Questions (Exact Formats)

Instructions

Вопросы сохранены в том же формате, как были на скринах: multiple choice, short answer, fill in blanks.

1. Fill in blanks

```
from rest_framework import generics

class ProductListAPIView(generics._____):
    queryset = Product.objects.all()
    serializer_class = ProductSerializer

class ProductDetailAPIView(generics._____):
    queryset = Product.objects.all()
    serializer_class = ProductSerializer
    _____ = 'product_id'
```

2. Multiple choice

Inside `REST_FRAMEWORK` in `settings.py`, which key configures global authentication classes?

- A) `AUTHENTICATION_BACKENDS` B) `DEFAULT_PERMISSION_CLASSES` C) `DEFAULT_AUTHENTICATION_CLASSES` D) `AUTH_CLASSES`
-

3. Multiple choice

Which DRF permission class restricts access to authenticated users only?

- A) `AllowAny` B) `IsAuthenticated` C) `IsAdminUser` D) `DjangoModelPermissions`
-

4. Short answer

Which `manage.py` sub-command starts Django's built-in development web server?

5. Fill in blanks

```
class Category(models.Model):  
    name = models.CharField(max_length=100)  
  
    def _____(self):  
        return self._____
```

6. Multiple choice

What does `unique=True` on a Django model field do?

A) Hides the field in admin B) Encrypts the value automatically C) Makes it the primary key D) Adds a database UNIQUE constraint so duplicate values are not allowed

7. Fill in blanks

```
class ProductCreateView(generics.CreateAPIView):  
    queryset = Product.objects.all()  
    serializer_class = ProductSerializer  
    permission_classes = [IsAuthenticated]  
  
    def _____(self, serializer):  
        serializer.save(_____self._____.user)
```

8. Short answer

Which DRF router class auto-generates URL patterns for a registered ViewSet?

9. Multiple choice

How do you pass the request object into a serializer so it can be accessed via

`self.context['request']`?

A) `ProductSerializer(product, request=request)` B) `ProductSerializer(product, context={'request': request})` C) `ProductSerializer(product, self.request manually)` D) This is impossible in DRF

10. Short answer

With DRF's `OrderingFilter` enabled, which URL query parameter requests result ordering? (the part before `=`)

11. Fill in blanks

```
from rest_framework import viewsets, routers

class ProductViewSet(viewsets._____):
    queryset = Product.objects.all()
    serializer_class = ProductSerializer

router = routers._____()
router._____('products', ProductViewSet)
```

12. Short answer

Which Simple JWT class-based view returns an access + refresh token pair when given valid credentials?

13. Short answer

Which `manage.py` sub-command scaffolds a brand-new Django app inside the project?

14. Multiple choice

Which queryset expression returns products whose `price` is strictly greater than `100`?

- A) `Product.objects.filter(price__gt=100)` B) `Product.objects.filter(price>100)` C) `Product.objects.where(price__gt=100)` D) `Product.objects.filter(price__gte=100)`
-

15. Multiple choice

When credentials are invalid, what does Django's `authenticate()` function return?

- A) `AnonymousUser` B) `ValidationError` C) `None` D) `{}`
-

16. Multiple choice

Which Django model field is most appropriate for storing money values such as product price?

- A) `FloatField` B) `IntegerField` C) `DecimalField` with `max_digits` and `decimal_places`
D) `CharField`
-

17. Fill in blanks

```
class ProductSerializer(serializers.ModelSerializer):
    class Meta:
        model = Product
        fields = '__all__'

    def _____price(self, value):
        if value < 0:
            raise serializers._____(
                'Price must be non-negative'
            )
        return _____
```

18. Multiple choice

In a `ModelSerializer`, what does:

```
fields = '__all__'
```

mean?

- A) Includes every model field in serializer output B) Only includes required fields C) Includes fields except id D) Dynamically chooses random fields
-

19. Short answer

In a sample JWT header:

```
{"alg": "____", "typ": "JWT"}
```

which short string fills the blank?

20. Short answer

With DRF's `SearchFilter` enabled, which URL query parameter triggers the search? (the part before `=`)

21. Short answer

When writing a class-based view in `urls.py`, which class method is called to turn the class into a callable view?

`TokenObtainPairView._____()`

22. Fill in blanks

```
@_____(detail=_____, methods=['get'])
def featured(self, request):
    qs = Product.objects.filter(is_featured=True)
    serializer = self.get_serializer(qs, _____=True)
```

23. Fill in blanks

```
from django.db import _____

class Category(_____.Model):
    name = _____.CharField(max_length=300)

    class _____:
        verbose_name = 'Category'
        verbose_name_plural = '_____'
```

24. Fill in blanks

```
from rest_framework_simplejwt.views import (
    _____,
    TokenRefreshView,
)

urlpatterns = [
    path('api/token/', _____(), name='token_obtain_pair'),
```

```
path('api/token/refresh/', _____.as_view(), name='token_refresh'),  
]
```

25. Fill in blanks

```
from django.views.decorators.csrf import ____  
from django.http import JsonResponse  
import json  
  
@____  
def products_list(request):  
    if request.method == 'POST':  
        data = _____.loads(request.____)
```

26. Multiple choice

Which command sequence creates and applies database migrations?

A) migrate then makemigrations B) runserver then migrate C) makemigrations then migrate D) collectstatic then migrate

27. Short answer

Complete the condition:

```
if _____ == 'GET':
```

28. Short answer

Which Python special method controls how a model instance is shown in admin/shell?

29. Short answer

Which filter backend class is commonly used with `filterset_fields`?

30. Multiple choice

What does `read_only_fields` mean in a serializer?

A) Hidden from output B) Included in output but ignored on input C) Required on POST only D) Only editable by admin

31. Multiple choice

Which HTTP method performs a full resource update?

A) POST B) PATCH C) UPDATE D) PUT

32. Multiple choice

What is the main purpose of CORS?

A) Database backups B) Control whether browsers allow one origin to access another origin C) Encrypt request body D) Generate JWT tokens

33. Multiple choice

Which class is the base class for Django models?

A) models.Schema B) django.orm.Entity C) models.Model D) django.db.Database

34. Multiple choice

Which two attributes do GenericAPIView mixins rely on?

A) queryset and serializer_class B) auth_classes and permission_classes C) renderer_classes and parser_classes D) model and fields

35. Multiple choice

What is the main purpose of `perform_create()`?

A) Inject extra data during save (e.g. owner=request.user) B) Generate URLs C) Send emails after response D) Replace validation

36. Multiple choice

What does `serializer.is_valid(raise_exception=True)` do on invalid input?

A) Raises ValidationError and returns HTTP 400 B) Enables strict mode only C) Returns False silently D) Stores errors only

37. Short answer

Which top-level dictionary in `settings.py` stores DRF-wide configuration?

38. Short answer

Which HTTP method returns supported methods for a URL?

39. Multiple choice

With DRF SearchFilter enabled, which URL triggers search for text `phone`?

- A) `http://url?filter=phone` B) `http://url?search=phone` C) `http://url?find=phone` D) `http://url?q=phone`
-

40. Fill in blanks

```
INSTALLED_APPS = [  
    'rest_framework',  
    '_____',  
]  
  
MIDDLEWARE = [  
    'corsheaders._____._____',  
]
```

41. Python coding

Read integer `n` and print `n!`.

42. Python coding

Read integers `a` and `b` and print their greatest common divisor.

43. Multiple choice

Which package should be installed to use Simple JWT with Django REST Framework?

- A) `django-rest-framework-simplejwt` B) `django-jwt-token` C) `restframework-jwt-auth` D) `simple-jwt-django`
-

44. Multiple choice

How do you restrict access to a single APIView class to authenticated users?

- A) Set `require_login = True` B) Set `permission_classes = [IsAuthenticated]` as a class attribute on the APIView C) Decorate the class with `@login_required` D) Put `permissions = IsAuthenticated` inside `Meta`
-

45. Multiple choice

Which DRF permission class allows only authenticated users?

- A) `AllowAny` B) `IsAdminUser` C) `IsAuthenticated` D) `DjangoModelPermissions`
-

46. Fill in blanks

```
def ____price(self, value):  
    if value < 0:  
        raise serializers.____('Price must be non-negative')  
    return ____
```

47. Multiple choice

In a `ModelSerializer.Meta`, what is true about `fields` and `exclude`?

- A) You may set either `fields` or `exclude`, but not both B) You must always set both C) `exclude` is only for admin models D) `fields` works only with function-based views
-

48. Multiple choice

How do you pass the request object into a serializer so it can be accessed via

`self.context['request']`?

A) `ProductSerializer(product, request=request)` B) `ProductSerializer(product, context={'request': request})` C) `ProductSerializer(product, self.request manually)` D) This is impossible in DRF

49. Multiple choice

How should you raise a validation error inside a DRF serializer validation method?

A) Call `self.fail()` with no arguments B) Raise `serializers.ValidationError(message)` C) Return `False` D) Set `self.errors = message`

50. Short answer

Which HTTP method returns the supported HTTP methods for the given URL?

51. Short answer

Which top-level dictionary name in `settings.py` stores Django REST Framework configuration?

52. Fill in blanks

GenericAPIView with mixins: fill in the mixin combination for a list + create view.

```
from rest_framework import generics, mixins

class ProductListAPIView(
    mixins._____,
    mixins.CreateModelMixin,
    generics._____):

    queryset = Product.objects.all()
    serializer_class = ProductSerializer

    def get(self, request, *args, **kwargs):
        return self._____(request, *args, **kwargs)
```

53. Fill in blanks

DjangoFilterBackend with `filterset_class`.

```

from django_filters.rest_framework import _____
from rest_framework.permissions import IsAuthenticated
from rest_framework import generics

class ProductListAPIView(generics.ListAPIView):
    queryset = Product.objects.all()
    serializer_class = ProductSerializer
    filter_backends = (_____,)
    _____ = ProductFilter
    _____ = (IsAuthenticated,)

```

54. Fill in blanks

Pip workflow: save and reinstall dependencies.

```

pip _____ > _____
pip _____ -r _____

```

55. Multiple choice

For a long, free-form description with no enforced length limit, which Django model field is most appropriate?

- A) `SlugField` B) `CharField` C) `BinaryField` D) `TextField`

56. Multiple choice

If `Product.category = models.ForeignKey(Category, related_name='products')`, how do you get all products belonging to category instance `cat`?

- A) `cat.products.all()` B) `Product.objects.get(cat)` C) `cat.product_set.fetch()` D) `cat.related(Product)`

57. Python coding

Read a single line of lowercase words separated by spaces. Print each unique word with its frequency, one per line, in the form `word count`, sorted alphabetically by word.

58. Python coding

Read one line containing a sentence with words separated by spaces. Print the number of words on the first line and the total number of non-space characters on the second line.

59. Fill in blanks

Custom `PageNumberPagination`.

```
from rest_framework._____ import PageNumberPagination

class StandardResultsPagination(_____):
    page_size = 10
    page_size_query_param = 'page_size'
    max_page_size = 100

class ProductListAPIView(generics.ListAPIView):
    queryset = Product.objects.all()
    serializer_class = ProductSerializer
    _____ = StandardResultsPagination
```

60. Fill in blanks

Cross-field validation with `validate()`.

```
class TimeRangeSerializer(serializers.Serializer):
    start = serializers.IntegerField()
    end = serializers.IntegerField()

    def _____(self, attrs):
        if attrs['start'] >= attrs['end']:
            raise serializers._____('start must be before end')
        return _____
```

Answers

1. `ListCreateAPIView`, `RetrieveUpdateDestroyAPIView`, `lookup_field`
2. C
3. B
4. `runserver`
5. `__str__`, `name`
6. D
7. `perform_create`, `owner=`, `request`

8. `DefaultRouter`
9. B
10. `ordering`
11. `ModelViewSet`, `DefaultRouter`, `register`
12. `TokenObtainPairView`
13. `startapp`
14. A
15. C
16. C
17. `validate`, `ValidationError`, `value`
18. A
19. `HS256`
20. `search`
21. `as_view`
22. `action`, `False`, `many`
23. `models`, `models`, `models`, `Meta`, `Categories`
24. `TokenObtainPairView`, `TokenObtainPairView`, `as_view`, `TokenRefreshView`
25. `csrf_exempt`, `csrf_exempt`, `json`, `body`
26. C
27. `request.method`
28. `__str__`
29. `DjangoFilterBackend`
30. B
31. D
32. B
33. C
34. A
35. A
36. A
37. `REST_FRAMEWORK`
38. `OPTIONS`
39. B
40. `corsheaders`, `middleware`, `CorsMiddleware`
41. `n=int(input()); r=1; for i in range(1,n+1): r*=i; print(r)`
42. `a=int(input()); b=int(input()); while b!=0: a,b=b,a%b; print(a)`
43. A
44. B
45. C
46. `validate`, `ValidationError`, `value`
47. A
48. B
49. B
50. `OPTIONS`
51. `REST_FRAMEWORK`
52. `ListModelMixin`, `GenericAPIView`, `list`
53. `DjangoFilterBackend`, `DjangoFilterBackend`, `filterset_class`, `permission_classes`
54. `freeze`, `requirements.txt`, `install`, `requirements.txt`
55. D
56. A

```
57. words=input().split(); used=[]; for w in sorted(words): if w not in used:  
    print(w, words.count(w)); used.append(w)  
58. line=input(); words=line.split(); print(len(words));  
    print(len(line.replace(' ', '')))  
59. pagination, PageNumberPagination, pagination_class  
60. validate, ValidationError, attrs
```